Structural Accessibility Boundaries and Recoverability in a Minimal H–O Redox Network Model

(Companion to rate-dependent recoverability dynamics)

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Abstract

We introduce a minimal network-based model to study accessibility and recoverability within a simplified H–O redox state space. Oxygen-containing species are represented as nodes connected by proton transfer and redox transitions, with edge weights encoding an abstract and heuristic transition cost.

We define recoverability operationally as the existence of a finite-cost path from a given state to a reference core state (H_2O). By introducing a threshold on allowable transition cost, the system appears to exhibit a transition from full accessibility to partial exclusion, and ultimately to a restricted core regime.

Within the present model configuration, ozone (O_3) appears as the first species to become inaccessible under tightening constraints. This behavior remains unchanged after removal of an intermediate node, suggesting that recoverability may not depend solely on individual pathways.

We further show that this boundary appears to be controlled by the cost of a specific return pathway ($O_3 \rightarrow O_2$), with a clear transition emerging at a critical value. This suggests that accessibility boundaries may arise from localized bottlenecks rather than purely global network structure.

The framework is intentionally minimal and should be interpreted as a conceptual and falsifiable basis for studying accessibility-limited behavior in redox-related systems.

1. Introduction

Failure in complex systems is often interpreted as the result of exceeding absolute limits such as energy or load. However, systems may also fail when transitions between states become inaccessible within finite constraints, even without any explicit violation of physical limits.

Recent work in dynamical systems has highlighted the role of recovery limitations and rate-dependent transitions, where system behavior depends not only on total input but also on the temporal structure of forcing. In parallel, chemical and biological systems may be viewed as networks of accessible states connected by transformation pathways.

However, systems may also fail when transitions between states become inaccessible within finite constraints, even without any explicit violation of physical limits. In particular, it becomes relevant to ask whether all states remain accessible to a reference equilibrium under bounded transition costs.

Here, we explore whether recoverability can be formulated as a structural property of a state-space network. Rather than focusing on concentrations or kinetics, we study whether states

remain accessible to a reference core under constrained transition cost, using shortest-path metrics in a minimal H–O redox network.

Building on this perspective, we consider the possibility that recoverability may be governed by a structural accessibility boundary in weighted path space, as formulated in the working hypothesis below.

Working hypothesis

This structural framework serves as a network-based counterpart to our previous minimal dynamical model (Ojeda, 2026), where recoverability was found to depend on both forcing rate and total load **(DOI: https://doi.org/10.5281/zenodo.19069265).**

We propose that recoverability in a minimal H–O transition network may be governed by a structural accessibility boundary in weighted path space.

Specifically, nodes whose minimal weighted distance to a designated core exceeds a threshold may become operationally non-recoverable.

This hypothesis would be falsified if accessibility patterns were found to be invariant under systematic perturbations of network weights or topology, indicating that no well-defined accessibility boundary exists.

2. Model

2.1 State space

We define a minimal set of oxygen-containing species:

- H_2O (reference core)
- H_3O^+, OH^-
- O_2
- O_2^-
- HO_2
- H_2O_2
- OH•
- O_3

These species are connected through a simplified set of allowed transformations, including proton transfer and redox processes. The goal is not to capture full chemical realism, but to define a minimal state space in which transitions between oxygen-related species can be represented.

2.2 Network representation

The system is modeled as a weighted directed graph:

- Nodes represent chemical species
- Edges represent allowed transitions between species
- Edge weights encode an abstract transition cost

These costs are not intended to represent exact thermodynamic or kinetic quantities. Instead, they provide a heuristic measure of the relative difficulty of moving between states within the network.

The network is not intended to represent a quantitatively accurate chemical system, but rather a minimal topological abstraction capturing possible connectivity between oxygen-related states.

2.3 Recoverability definition

For each node we compute the minimum weighted path cost to a reference core state, taken here as H_2O.

Recoverability is defined operationally as:

- recoverable if the minimum path cost ≤ threshold
- non-recoverable otherwise

This defines a threshold-dependent accessibility boundary within the state space. The threshold parameter is treated as an abstract constraint on transition cost and is not assigned direct physical units.

By varying the threshold, we explore how the set of recoverable states changes as constraints on transition cost are tightened or relaxed.

3. Results

3.1 Accessibility structure

We first analyze the accessibility of states as a function of the allowed transition cost. For each node, we compute the minimum weighted path to the core state (H_2O) and classify it as recoverable or non-recoverable based on a threshold.

At low threshold (threshold = 1), only core species remain accessible, while all peripheral states become non-recoverable. This defines a highly restrictive regime in which only direct or near-equilibrium transitions are permitted.

At high threshold (threshold = 3), all states become accessible, corresponding to a permissive regime in which transition cost does not limit accessibility.

3.2 Intermediate regime (threshold = 2)

At an intermediate threshold (threshold = 2), a selective boundary emerges. Most species remain recoverable, while ozone (O_3) appears as non-recoverable within the present model configuration.

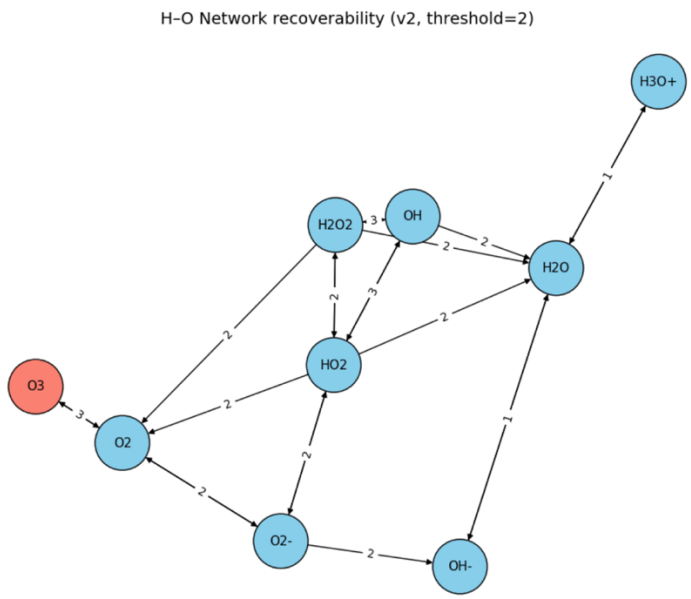

Figure 1. *Recoverability classification in the H–O transition network (v2, threshold = 2). Nodes are classified as recoverable or non-recoverable based on the minimum weighted path cost to the core (H_2O).*

As shown in Figure 1, this result is consistent with the emergence of a selective accessibility boundary at intermediate threshold.

This suggests that O_3 may occupy a structurally more distant region of the state space, requiring higher-cost transitions to return to the core.

This separation appears to emerge from the network structure and assigned transition costs, rather than being imposed explicitly.

3.3 Robustness to topology

To test whether recoverability depends on specific pathways, we remove an intermediate node (HO_2) and recompute accessibility.

As shown in Figure 2, the overall classification appears unchanged: O_3 continues to appear as non-recoverable at threshold = 2, while other species remain recoverable. This suggests that the accessibility boundary is robust to removal of individual intermediate nodes.

Importantly, this stability is observed despite removal of a highly connected intermediate node.

The weighted accessibility to the core also remains unchanged across network variants (Figure 2), indicating that minimal return costs are insensitive to removal of individual intermediate nodes.

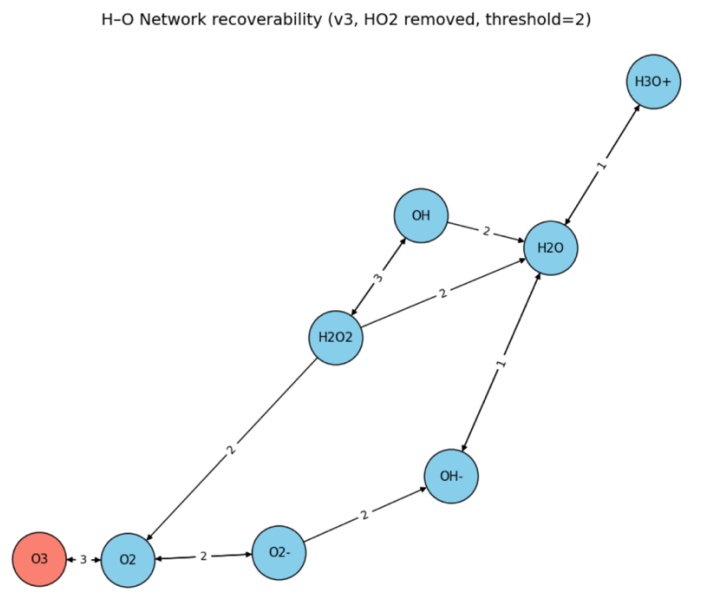

Figure 2. *Recoverability classification in the reduced H–O transition network (v3, HO_2 removed, threshold = 2). The recoverability structure remains qualitatively unchanged after removal of the HO_2 intermediate, suggesting robustness to changes in network topology.*

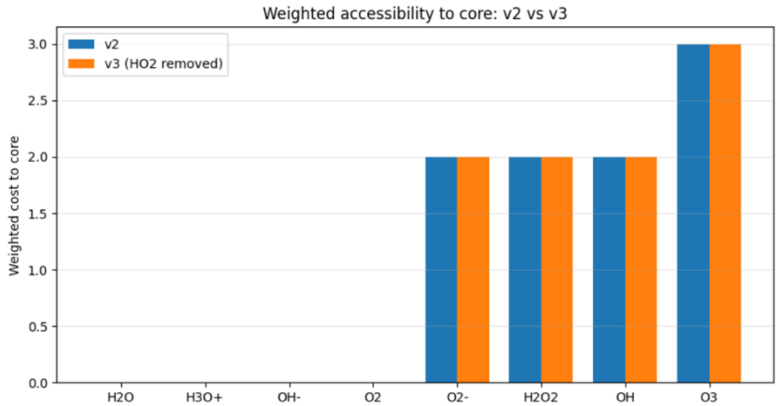

Figure 3. *Weighted accessibility to the core (H2O) in v2 and v3 networks.*

Minimal path costs remain unchanged after removal of the HO2 intermediate, indicating that recoverability classification is governed by global network structure rather than specific pathways.

These results are consistent with the possibility that recoverability may be influenced by global network structure rather than by the presence of specific transitions.

This behavior suggests that the accessibility boundary reflects a global property of the network rather than a local feature of individual pathways.

Importantly, this stability is observed despite removal of a highly connected intermediate node.

To further characterize this global behavior, we examine how recoverability classification varies with the allowed transition cost.

As shown in Figure 4, the number of non-recoverable species decreases as the threshold increases, indicating a transition between distinct accessibility regimes. At low threshold, only core states remain accessible. At intermediate threshold, selective exclusion appears. At high threshold, all states become accessible.

Within the present model configuration, O_3 appears as the first species to become excluded under tightening constraints.

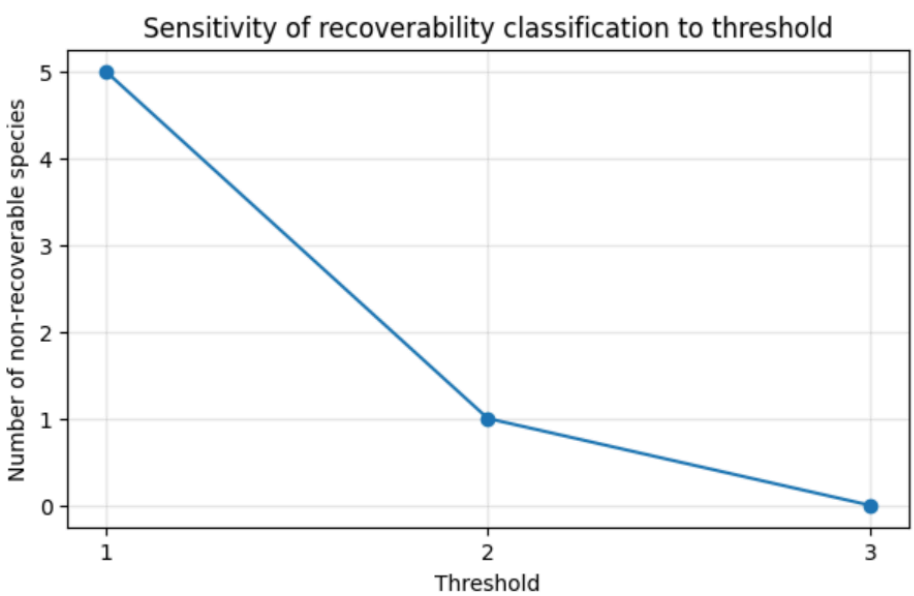

Figure 4. *Sensitivity of recoverability classification to threshold.*
The number of non-recoverable species decreases as the allowed transition cost increases, indicating a transition between distinct accessibility regimes. Ozone (O_3) appears as the first species to become excluded under tightening constraints within the present model configuration

3.4 Robustness to weight perturbations

To test whether the observed accessibility boundary depends on a specific choice of edge weights, we performed a perturbation analysis in which all transition weights were randomly varied within ±20% of their baseline values.
Across 200 independent realizations, the classification of O_3 as non-recoverable remained invariant, while intermediate species showed partial sensitivity to weight variation.

Notably, O_3 exceeded the accessibility threshold in all realizations, indicating that its separation from the core is not dependent on fine-tuning of weights.

These results suggest that the accessibility boundary reflects a structural property of the network rather than an artifact of a particular weighting scheme.

This separation is visualized in Figure 5, where O_3 remains consistently beyond the accessibility threshold across perturbations.

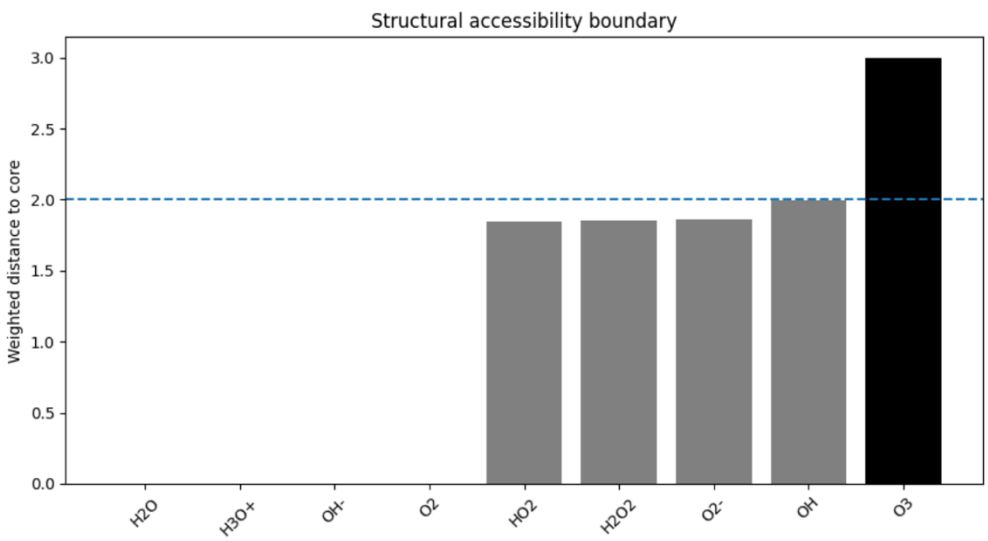

Figure 5. *Structural accessibility boundary. Mean weighted distance of each species to the core (H_2O) across weight perturbations, compared against the recoverability threshold (dashed line). O_3 consistently lies beyond the threshold, indicating a stable separation from the recoverable regime*

4. Discussion

The results presented here suggest that recoverability in a minimal H–O transition network may be governed by an accessibility boundary defined in weighted path space.

Across both topology modifications and weight perturbations, the qualitative classification of species remains largely stable, with O_3 consistently appearing beyond the recoverable regime under the tested conditions.

This behavior indicates that recoverability may not depend solely on total load or individual transitions, but rather on the global structure of the network and the accessibility of return pathways to the core.

While the present model is intentionally minimal, the observed robustness of the accessibility boundary suggests consistency with broader classes of systems in which recovery depends on pathway accessibility rather than solely on magnitude of perturbation.

In this sense, the accessibility boundary identified here may represent a simplified abstraction of systems where functional recovery is constrained by network structure and transition costs.

Further work would be required to determine whether analogous accessibility constraints emerge in biochemical or redox-regulated systems.

5. Connection to dynamical systems

The structural framework introduced here may be viewed as complementary to dynamical models of recovery-limited systems.

In dynamical formulations, system behavior is often characterized in terms of time-dependent responses, such as whether a system returns to a baseline state after perturbation, and how recovery depends on the rate and duration of forcing. In such models, failure may arise when recovery processes cannot keep pace with imposed demand.

In contrast, the present model focuses on the accessibility of return paths within a state-space representation. Rather than modeling time evolution explicitly, we consider whether a return path exists within a bounded transition cost.

Taken together, these perspectives suggest a possible connection between dynamical and structural constraints. In particular, the existence of a path in state space does not guarantee recoverability if the cost associated with that path exceeds the system's effective capacity.

In this sense, failure may be interpreted as arising not from the absence of return paths, but from the inaccessibility of those paths under finite constraints. This interpretation is consistent with recovery-limited dynamics, where system behavior depends on whether internal processes can operate within the required timescales.

This correspondence suggests that structural accessibility constraints may provide a minimal representation of underlying dynamical limitations.

6. Controlled perturbation of return-path cost

6.1 Identification of a dominant edge

Edge-level analysis suggests that not all transitions contribute equally to the observed accessibility structure.

In particular, the transition $O_3 \rightarrow O_2$ shows a strong association with the accessibility of O_3 across perturbations. This suggests that the return pathway from O_3 to the core may play a disproportionate role in determining whether O_3 remains within the recoverable regime.

6.2 Directed perturbation experiment

To test this hypothesis, we performed a controlled perturbation experiment in which all edge weights were held fixed except for the $O_3 \rightarrow O_2$ transition.

The weight of this edge was varied continuously across a broad range, while recomputing the weighted distance to the core for all species.

This approach isolates the effect of a single transition, allowing us to distinguish between global network effects and localized control mechanisms.

6.3 Response of accessibility thresholds

To evaluate the effect of localized perturbations, we analyzed how accessibility thresholds vary under controlled changes to the cost of the $O_3 \rightarrow O_2$ transition, while keeping all other edge weights fixed.

For each value of the return-path cost, we computed the minimum weighted distance from each species to the core. This provides an operational measure of accessibility within the constrained network.

As shown in Figure 6, the accessibility threshold of O_3 increases approximately linearly with the cost of the $O_3 \rightarrow O_2$ transition. In contrast, the most distant intermediate species remains largely unaffected across the same range.

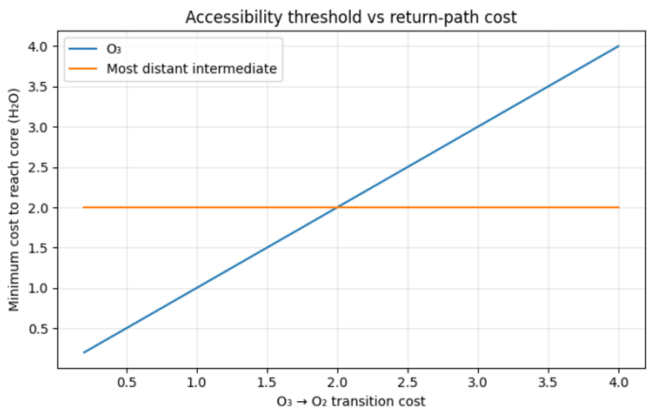

Figure 6. Accessibility thresholds as a function of the $O_3 \rightarrow O_2$ transition cost. The minimum cost required for O_3 to reach the core increases approximately linearly with the cost of the return pathway, while the most distant intermediate remains stable. This indicates that accessibility is controlled by a localized bottleneck.

This differential response suggests that accessibility is not uniformly distributed across the network, but instead may be governed by specific transitions that act as bottlenecks.

In particular, the $O_3 \rightarrow O_2$ pathway appears to control the ability of O_3 to remain connected to the core under increasing constraints, indicating a localized mechanism underlying the observed accessibility structure.

6.4 Emergence of a critical boundary

To quantify the emergence of separation, we define the gap between O_3 and the most distant intermediate species.

As shown in Figure 7, this gap crosses zero at a well-defined critical value of the $O_3 \rightarrow O_2$ weight (~2).

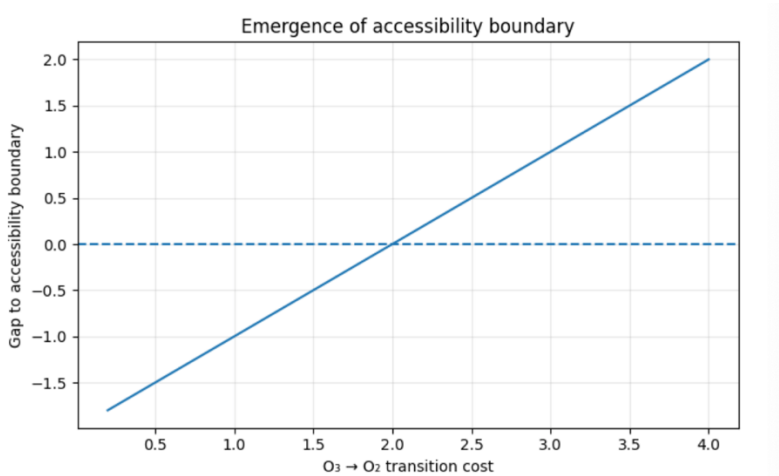

Figure7. *Emergence of an accessibility boundary. The gap between O_3 and the most distant intermediate crosses zero at a critical value of the $O_3 \rightarrow O_2$ transition cost (~2). This marks the transition from an integrated to a separated accessibility regime.*

Below this value, O_3 remains integrated within the accessibility structure. Above it, O_3 becomes separated from the recoverable regime.

This behavior is consistent with the emergence of a regime-dependent boundary controlled by a localized bottleneck.

6.5 Interpretation

These results suggest that the accessibility boundary is not an intrinsic topological feature of the network.

Instead, it appears to emerge when the cost of the return pathway exceeds the characteristic accessibility scale of the system.
This supports a bottleneck-controlled interpretation, in which specific transitions can regulate global accessibility structure.

7. Limitations

The model presented here is intentionally minimal and does not aim to capture the full complexity of real chemical or biological systems.

First, the transition costs assigned to edges are heuristic and are not derived from experimentally measured thermodynamic or kinetic parameters. The resulting accessibility structure may therefore depend on the weighting scheme chosen.

Second, the model does not include explicit kinetics, concentrations, or time-dependent dynamics. It therefore does not describe how systems evolve in time, but only whether transitions between states are structurally accessible under a given constraint.

Third, the threshold parameter is a free and abstract quantity without direct physical units. While qualitative patterns appear across multiple threshold values, the exact classification of states depends on this parameter.

Finally, the network itself represents a simplified subset of oxygen-related species and transitions. Additional species, alternative pathways, or different weighting schemes may alter the structure of the accessibility landscape.

For these reasons, the present work should be interpreted as a conceptual and computational framework intended to explore accessibility-limited behavior, rather than as a quantitatively validated chemical model.

8. Conclusion

We have presented a minimal network-based framework to study accessibility and recoverability within a simplified H–O redox state space.

The results suggest that the system appears to exhibit a threshold-dependent accessibility structure, in which the set of recoverable states depends on the allowed transition cost. As this constraint is varied, the system transitions from a fully accessible regime to a partially restricted regime, and ultimately to a core-only regime.

Within the present model configuration, ozone (O_3) appears as the first species to become inaccessible as constraints tighten. While this behavior remains stable under removal of an intermediate node, further analysis indicates that the accessibility boundary is not purely a global topological property.
Instead, the boundary appears to emerge when the cost of the return pathway ($O_3 \rightarrow O_2$) exceeds a characteristic accessibility scale of the system. This suggests a bottleneck-controlled, regime-dependent transition governed by specific transitions rather than by the network structure alone.

More generally, this work supports the idea that system behavior may depend on accessibility within a state space rather than on individual species alone. In this sense, failure may arise when return paths exist but are no longer accessible within the effective constraints of the system.

While the model is intentionally minimal, it provides a simple and falsifiable framework for exploring accessibility-limited behavior and its possible connection to recovery-limited dynamics.

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